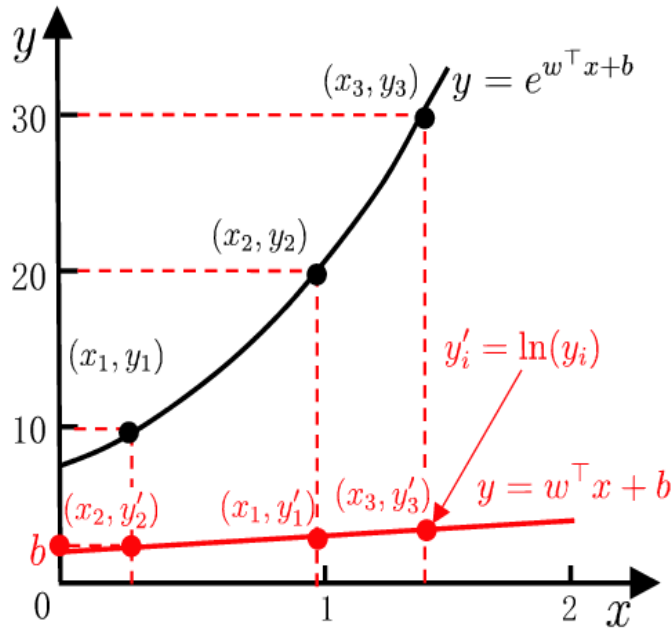


Log-Linear Regression

非线性变化来构建线性模型

- The logarithm of the output label can be used for approximation



$$\ln y = w^T x + b$$



$$y = w^T x + b$$

Linear Regression - Generalized Linear Model

- The general form

$$y = g^{-1}(\mathbf{w}^T \mathbf{x} + b)$$

- Where the function $g(\cdot)$ is the link function.
 - a monotonic differentiable function $g(\cdot) = \ln(\cdot)$
- Log-linear regression is a special case of generalized linear models when

Feature mappings

- We get polynomial regression for free!

- Define the feature map
$$\phi(x) = \begin{pmatrix} 1 \\ x \\ x^2 \\ x^3 \end{pmatrix}$$

- Polynomial regression model:
$$y = \sum_{j=0}^n w_j \phi_j(x)$$

- All of the derivations and algorithms so far in this lecture remain exactly the same!

Non-Linear basis function

- So far we only used the observed values $x_1, x_2, \dots$
- However, linear regression can be applied in the same way to **functions** of these values
 - E.g.: add a new variable $z = x_1 x_2$ so each example becomes: $x_1, x_2, \dots, z$
- As long as these functions can be directly computed from the observed values the parameters are still linear in the data and the problem remains a multi-variate linear regression problem

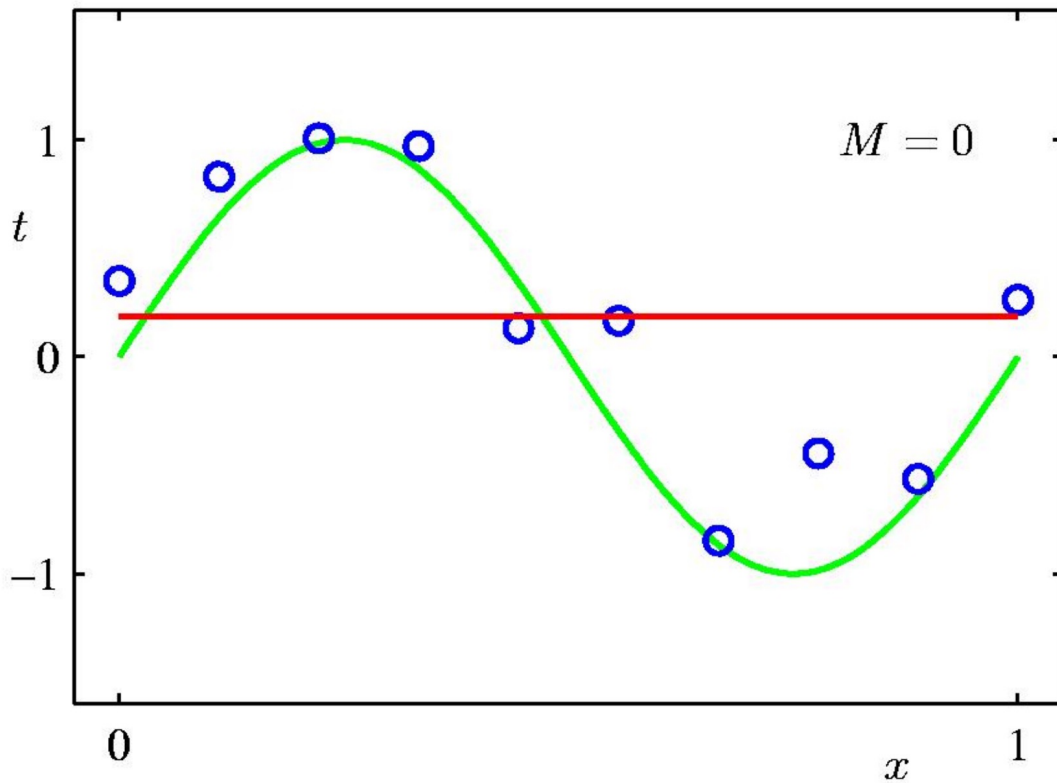
$$y = b + w_1 x_1^2 + \dots + w_d x_d^2 + \epsilon$$

Any function of the input values can be used. The solution for the parameters of the regression remains the same.

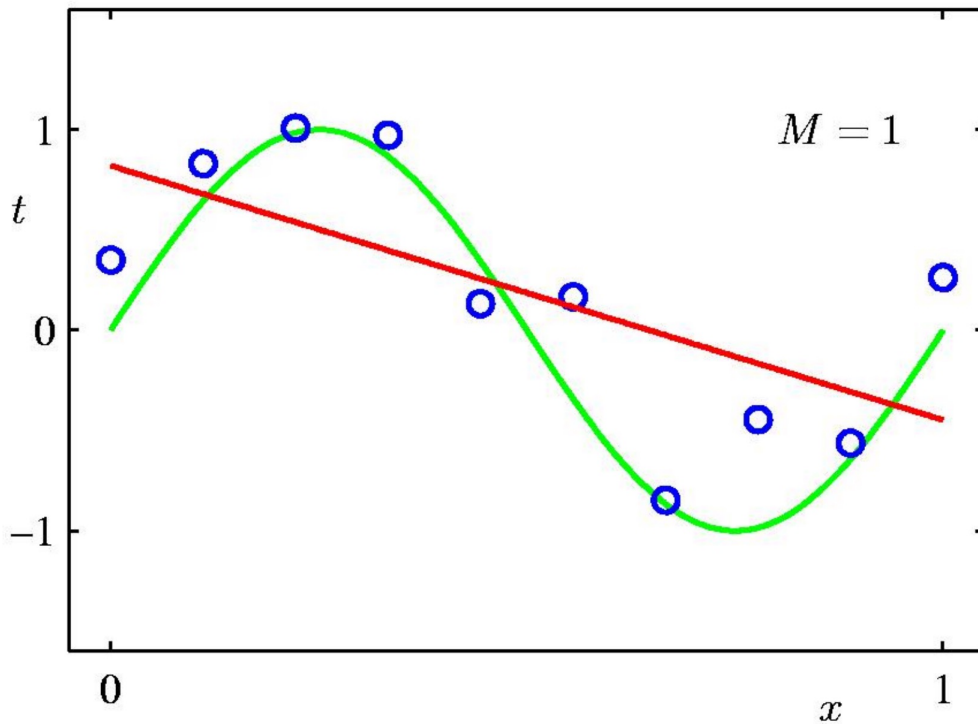
An example: polynomial basis vectors on a small dataset

– From Bishop Ch 1

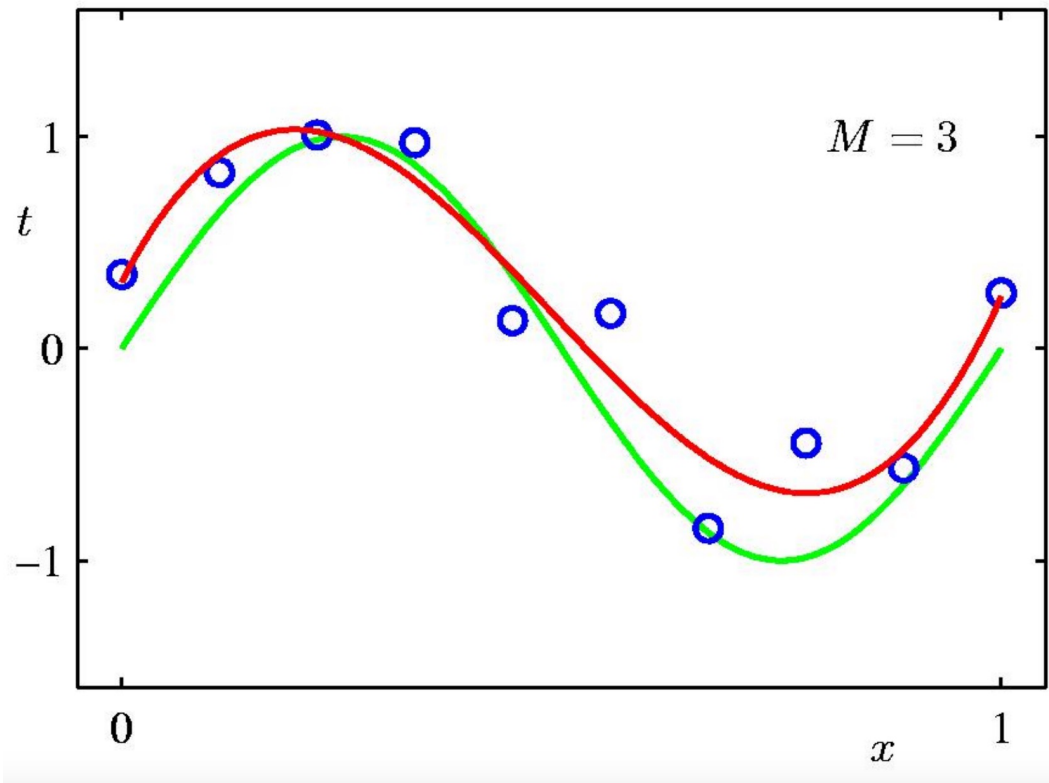
0th Order Polynomial



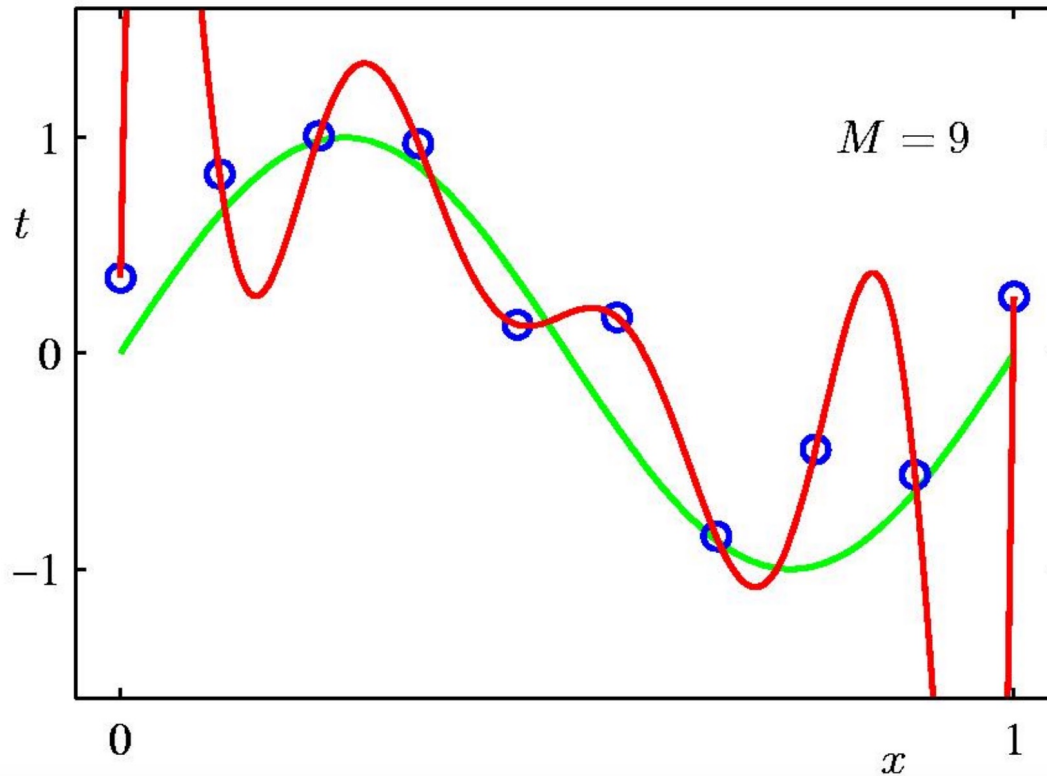
1st Order Polynomial



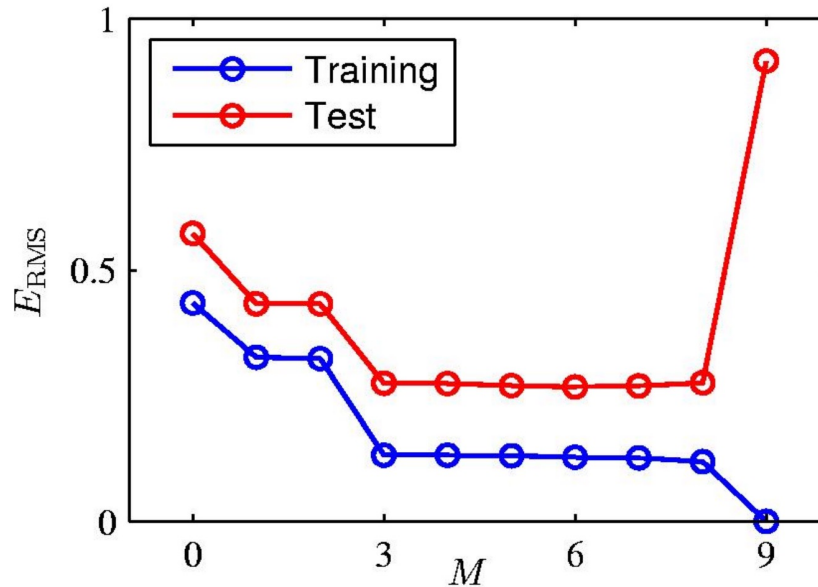
3rd Order Polynomial



9th Order Polynomial



Overfitting



Root-Mean-Square (RMS) Error

Polynomial Coefficients

为了完美地穿过每一个训练数据点（包括噪声），模型需要产生非常“陡峭”和“曲折”的曲线。在数学上就体现为高阶项拥有非常大的权重，因为只有大的权重才能产生这种剧烈变化的函数形态。

	$M = 0$	$M = 1$	$M = 3$	$M = 9$
w_0^*	0.19	0.82	0.31	0.35
w_1^*		-1.27	7.99	232.37
w_2^*			-25.43	-5321.83
w_3^*			17.37	48568.31
w_4^*				-231639.30
w_5^*				640042.26
w_6^*				-1061800.52
w_7^*				1042400.18
w_8^*				-557682.99
w_9^*				125201.43

Regularization

正则化:
Regularizer: a function that quantifies how much we prefer one hypothesis vs. another

Ridge regression and *Lasso*

- We can fit a model containing all features using a technique that constrains or regularizes the coefficient estimates, or equivalently, that **shrinks the coefficient estimates towards zero**.
- It may not be immediately obvious why such a constraint should improve the fit, but it turns out that shrinking the coefficient estimates can significantly reduce their variance.

Ridge Regression 岭回归

- Recall that the least squares fitting procedure estimates $\beta_0, \beta_1, \dots, \beta_p$ using the values that minimize

$$J(\boldsymbol{\beta}) = \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2$$

- In contrast, the ridge regression coefficient estimates are the values that minimize

$$J_{RR}(\boldsymbol{\beta}) = \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p \beta_j^2$$

What's the solution?

$\lambda \|\boldsymbol{\beta}\|_2^2$

where $\lambda \geq 0$ is a *tuning parameter*, to be determined separately.

Ridge Regression

- As with least squares, ridge regression seeks coefficient estimates that fit the data well.
- However, the second term, $\lambda \sum_j \beta_j^2$, called a *shrinkage penalty*, is small when $\beta_0, \beta_1, \dots, \beta_p$ are close to zero, and so it has the effect of *shrinking* the estimates of β_j towards zero.
- The tuning parameter λ serves to control the relative impact of these two terms on the regression coefficient estimates.
- Selecting a good value for λ is critical; cross-validation is used for this.

Ridge Regression

MAP与RR的关系.

Bayesian interpretation: MAP estimation with a **Gaussian prior** on the parameters

$$\beta^{MAP} = \operatorname{argmax}_{\beta} \sum_{i=1}^m \log p_{\beta}(y^{(i)} | \mathbf{x}^{(i)}) + \log p(\beta)$$

$$= \operatorname{argmin}_{\beta} J_{RR}(\beta)$$

where $p(\beta) \sim \mathcal{N}(0, \frac{1}{\lambda})$

$p(\beta | y, x) \propto p(y | x, \beta) p(\beta)$

Ridge Regression

- Recall the gradient descent update:

$$\beta^{t+1} \leftarrow \beta^t - \alpha \frac{\partial \mathcal{J}}{\partial \beta}$$

- The gradient descent update of the regularized cost has an interesting interpretation as **weight decay**:

$$\begin{aligned}\beta^{t+1} &\leftarrow \beta^t - \alpha \left(\frac{\partial \mathcal{J}}{\partial \beta} + \lambda \frac{\partial \mathcal{R}}{\partial \beta} \right) \\ &= \beta^t - \alpha \left(\frac{\partial \mathcal{J}}{\partial \beta} + \lambda \beta^t \right) \\ &= (1 - \alpha \lambda) \beta^t - \alpha \frac{\partial \mathcal{J}}{\partial \beta}\end{aligned}$$

The Lasso

- Ridge regression does have one obvious disadvantage: ridge regression will include all features in the final model
- The Lasso is a relatively recent alternative to ridge regression that overcomes this disadvantage. The lasso coefficients minimize the quantity

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j|$$

What's the solution?

 $\lambda ||\boldsymbol{\beta}||_1$

- In parlance, the lasso uses an ℓ_1 (pronounced “ell 1”) penalty statistical instead of an ℓ_2 penalty.

Optimization for LASSO

- The L_1 penalty is subdifferentiable (i.e. not differentiable at 0, 高维空间中的非光滑函数)
- An array of optimization algorithms exist to handle this issue:
 - Coordinate Descent (坐标下降法, 转化为一系列单变量子问题)
 - Block coordinate Descent (Tseng & Yun, 2009)
 - Sparse Reconstruction by Separable Approximation (SpaRSA) (Wright et al., 2009)
 - Fast Iterative Shrinkage Thresholding Algorithm (FISTA) (Beck & Teboulle, 2009)

The Lasso

- As with ridge regression, the lasso shrinks the coefficient estimates towards zero.
- However, in the case of the lasso, the ℓ_1 penalty has the effect of forcing some of the coefficient estimates to be exactly equal to zero when the tuning parameter λ is sufficiently large.
- Hence, much like best subset selection, the lasso performs *feature selection*.
- We say that the lasso yields *sparse* models — that is, models that involve only a subset of the variables.
- As in ridge regression, selecting a good value of λ for the lasso is critical; cross-validation is again the method of choice.

The Lasso

- Bayesian interpretation: MAP estimation with a **Laplace prior** on the parameters

$$\begin{aligned}\boldsymbol{\beta}^{MAP} &= \operatorname{argmax}_{\boldsymbol{\beta}} \sum_{i=1}^m \log p_{\boldsymbol{\beta}}(y^{(i)} \mid \boldsymbol{x}^{(i)}) + \log p(\boldsymbol{\beta}) \\ &= \operatorname{argmin}_{\boldsymbol{\beta}} J_{RR}(\boldsymbol{\beta})\end{aligned}$$

where $p(\boldsymbol{\beta}) \sim \text{Laplace}(0, f(\lambda))$

The Feature Selection Property of the Lasso

Why is it that the lasso, unlike ridge regression, results in coefficient estimates that are exactly equal to zero?

One can show that the lasso and ridge regression coefficient estimates solve the problems

$$\underset{\beta}{\text{minimize}} \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 \quad \text{subject to} \quad \sum_{j=1}^p |\beta_j| \leq s$$

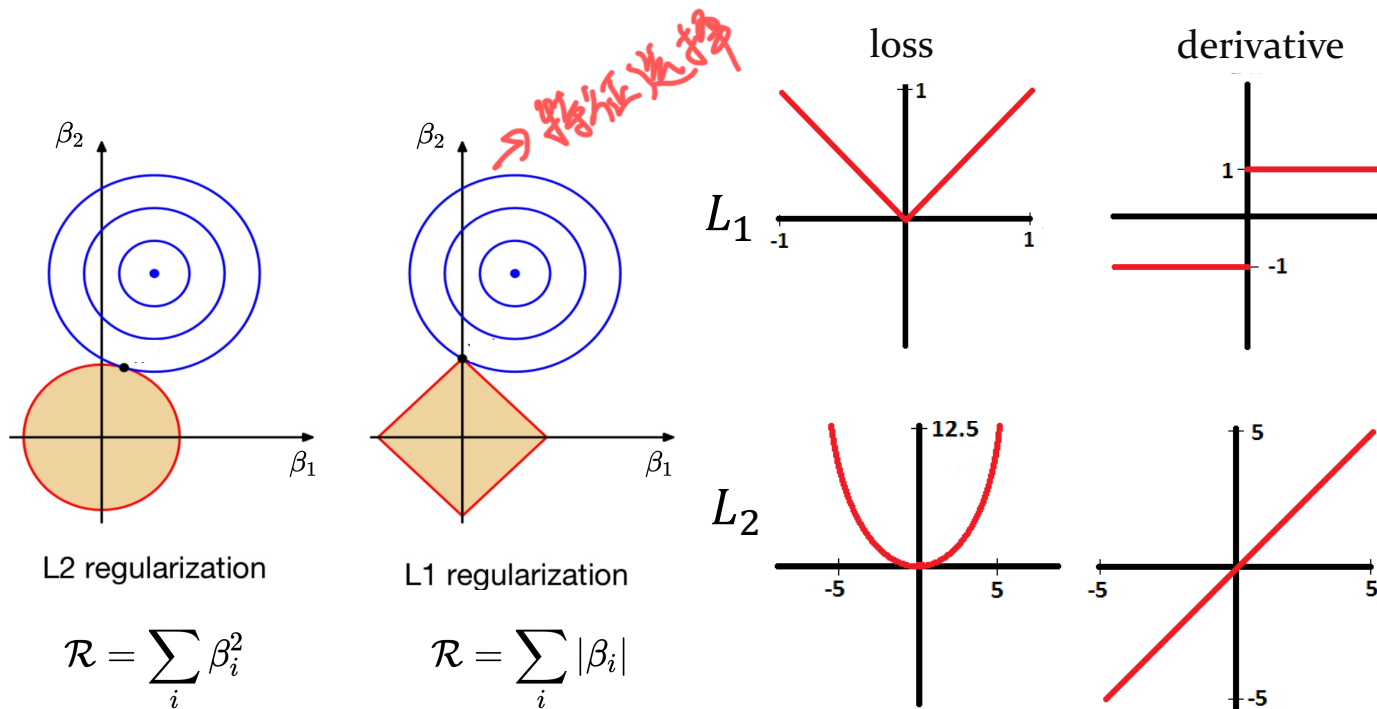
and

$$\underset{\beta}{\text{minimize}} \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 \quad \text{subject to} \quad \sum_{j=1}^p \beta_j^2 \leq s,$$

respectively.

Ridge Regression vs. Lasso

Why is it that the lasso, unlike ridge regression, results in coefficient estimates that are exactly equal to zero?



注：坐标下降法中，当参数估计值小于一个阈值 (λ) 时，优化算法会直接将其硬性设置为零

Regularization

- The linear model has distinct advantages in terms of its interpretability and often shows good predictive performance.
- **Model Interpretability:** By removing irrelevant features —that is, by setting the corresponding coefficient estimates to zero — we can obtain a model that is more easily interpreted. We will present some approaches for automatically performing *feature selection*.